

LESSON 5-3 • SESSION 2 • PRACTICE • FORM B

The Trapezoid Area Formula

Learning target – I can find the area of a trapezoid using the trapezoid area formula. • 6.G.A.1

REMEMBER

- $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The bases b_1 and b_2 are the two **parallel** sides.
- **Add** the bases first, multiply by the height, then take **half**. Slanted sides are not used.
- Missing base: double the area, divide by the height, then subtract the base you know.

WORKED EXAMPLE

Bases 6 m and 12 m, height 4 m.

Step 1 Add the bases: $6 + 12 = 18$.**Step 2** Multiply by the height: $18 \times 4 = 72$.**Step 3** Take half: $72 \div 2 = 36$.**Answer** $A = 36 \text{ m}^2$ **WORDS** base (base) • formula (fórmula) • right trapezoid (trapezio rectángulo)

PRACTICE

show your work • label units

1. Find the area of each trapezoid with bases
- b_1
- and
- b_2
- and height
- h
- .

a. 7 cm, 13 cm; $h = 6 \text{ cm}$ ____ cm^2

b. 14 in, 22 in; $h = 10 \text{ in}$ ____ in^2

c. 18 ft, 26 ft; $h = 5 \text{ ft}$ ____ ft^2

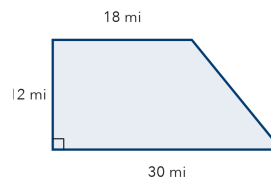
d. 4 m, 12 m; $h = 9 \text{ m}$ ____ m^2

- 2.
- County Land.**
- A county is shaped like this right trapezoid.

a. Sum of the bases ____ mi

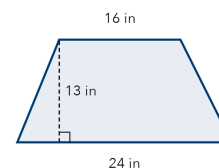
b. Area of the county ____ mi^2

c. Two copies make a rectangle. How long is it? ____ mi



- 3.
- Table Runner.**
- A table runner has the trapezoid shape shown. Find its area.

Show your work.

**Answer** _____ in^2

- 4.
- Sidewalk Panel.**
- A sidewalk panel is a trapezoid with parallel sides 6 m and 18 m and a height of 9 m. Find its area.

Show your work.

Answer _____ m^2

- 5.
- Marcus's Answer.**
- A trapezoid has bases 13 cm and 19 cm and a height of 15 cm. Marcus says its area is
- 480 cm^2
- . Is Marcus right?

Hint Check each step of the formula. Was a step skipped?

Show your work.

Answer _____ cm^2

- 6. Pool Cover.** Bases 18 ft and 26 ft, height 10 ft. Fabric costs \$3 per ft². Find the area and cost.
Hint Find the area first. Then multiply by the price.

Show your work.

Area, cost _____

- 7. Two Signs.** Sign M has bases 8 cm and 28 cm. Sign N has bases 12 cm and 24 cm. Both are 7 cm tall. Find each area. What do you notice?
Hint Compare the sums of the bases.

Show your work.

Notice _____

- 8. Missing Base.** A trapezoid has an area of 90 m² and a height of 6 m. One base is 11 m.
Hint Double the area, then divide by the height.
a. What is the sum of the two bases? _____ m
b. Find the other base. _____ m

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Andre’s Work.** A trapezoid has bases 9 m and 15 m and a height of 6 m. Andre wrote: “ $A = \frac{1}{2} \times 9 \times 6 = 27 \text{ m}^2$.” Explain the mistake and find the correct area.

TEST PRACTICE

choose the best answer

- 10.** A trapezoid has bases 4 in and 12 in and a height of 7 in. Its area is –
A 112 in² **B** 168 in² **C** 23 in² **D** 56 in²
- 11.** The trapezoid formula uses the average of the –
A two bases **B** two slanted sides
C height and one base **D** four sides

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

288 A	126 G	260 V	240 R	660 A	520 Z	480 Q	108 E	19 E	216 X
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____